

Customer Churn Prediction (Bank Customers)

1. Introduction and Problem Statement

Context: Customer retention is critical for the banking industry. It is far more expensive to acquire a new customer than to keep an existing one. This project focuses on **Churn Prediction**—identifying customers who are likely to close their accounts ("churn") based on their usage behavior and demographics.

Problem Statement: The objective is to build a classification model to predict whether a bank customer will exit (churn) or stay. We will process the **Churn Modelling Dataset**, handle categorical variables like Geography and Gender, and train a model to not only predict churn but also identify which features (e.g., Age, Balance, Credit Score) are the most significant indicators of a customer leaving.

Goals:

- Clean the dataset by removing non-predictive identifiers.
 - Encode categorical data (Gender, Geography) for machine learning.
 - Train a Random Forest Classifier.
 - Analyze and visualize **Feature Importance** to understand the drivers of churn.
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2. Loading the Dataset

- In this section we have loaded all required libraries for this project.
- Also we have downloaded the dataset and loaded it in our project.
- For the verification of dataset loaded successfully we have checked its first 5 rows by displaying the dataset

```
# Importing necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os
import kagglehub

# Scikit-learn modules
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

```
# =====
# 2. Loading the Dataset
# =====

# Download latest version of the dataset using KaggleHub
print("Downloading dataset...")
path = kagglehub.dataset_download("amisha0528/churn-modelling-
dataset")
print(f"Dataset downloaded to: {path}")

# Find the CSV file in the downloaded directory
csv_files = [f for f in os.listdir(path) if f.endswith('.csv')]
if csv_files:
    csv_path = os.path.join(path, csv_files[0])
    df = pd.read_csv(csv_path)
    print(f"Successfully loaded: {csv_files[0]}")
else:
    print("Error: No CSV file found in the downloaded path.")

# Display the first 5 rows
display(df.head())
```

Downloading dataset...

Dataset downloaded to: /kaggle/input/churn-modelling-dataset

Successfully loaded: Churn_Modelling.csv

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age
0	1	15634602	Hargrave	619	France	Female	42
1	2	15647311	Hill	608	Spain	Female	41
2	3	15619304	Onio	502	France	Female	42
3	4	15701354	Boni	699	France	Female	39
4	5	15737888	Mitchell	850	Spain	Female	43

	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	\
0	2	0.00	1	1	1	
1	1	83807.86	1	0	1	
2	8	159660.80	3	1	0	
3	1	0.00	2	0	0	
4	2	125510.82	1	1	1	

	EstimatedSalary	Exited
0	101348.88	1
1	112542.58	0
2	113931.57	1

3	93826.63	0
4	79084.10	0

3. Data Cleaning and Preparation

To prepare the data for modeling, we must remove columns that provide no predictive power.

- **RowNumber:** Just an index.
- **CustomerId:** Random unique identifier.
- **Surname:** Personal identifier (not useful for general prediction).

We will drop these columns and check for any missing values.

```
# Drop irrelevant columns
cols_to_drop = ['RowNumber', 'CustomerId', 'Surname']
df = df.drop(columns=cols_to_drop)
print(f"Dropped columns: {cols_to_drop}")

# Check for missing values
print("\nMissing Values Check:")
print(df.isnull().sum())

# Display the new structure
print(f"\nNew Dataset Shape: {df.shape}")

Dropped columns: ['RowNumber', 'CustomerId', 'Surname']

Missing Values Check:
CreditScore      0
Geography        0
Gender           0
Age              0
Tenure           0
Balance          0
NumOfProducts   0
HasCrCard        0
IsActiveMember   0
EstimatedSalary  0
Exited           0
dtype: int64

New Dataset Shape: (10000, 11)
```

4. Encoding Categorical Variables

Machine Learning models require numerical input. We have two key categorical features:

1. **Gender:** Binary (Male/Female). We will use **Label Encoding**.
2. **Geography:** Nominal (France, Spain, Germany). We will use **One-Hot Encoding** (creating dummy variables) to avoid assigning an arbitrary order to countries.

```
# 1. Label Encoding for Gender
label_encoder = LabelEncoder()
df['Gender'] = label_encoder.fit_transform(df['Gender'])
print("Gender encoded (0/1).")

# 2. One-Hot Encoding for Geography
# drop_first=True helps avoid multicollinearity (dummy variable trap)
df = pd.get_dummies(df, columns=['Geography'], drop_first=True)
print("Geography One-Hot Encoded.")

# Display the first few rows to see the changes
display(df.head())
```

Gender encoded (0/1).

Geography One-Hot Encoded.

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts
0	619	0	42	2	0.00	1
1						
1	608	0	41	1	83807.86	1
0						
2	502	0	42	8	159660.80	3
1						
3	699	0	39	1	0.00	2
0						
4	850	0	43	2	125510.82	1
1						

	IsActiveMember	EstimatedSalary	Exited	Geography_Germany
0	1	101348.88	1	False
False				
1	1	112542.58	0	False
True				
2	0	113931.57	1	False
False				
3	0	93826.63	0	False
False				
4	1	79084.10	0	False
True				

5. Model Training (Random Forest)

We will use a **Random Forest Classifier**.

- **Why Random Forest?** It generally provides high accuracy and, importantly for this task, it offers a built-in method to calculate **Feature Importance**, allowing us to see exactly which factors contribute most to customer churn.

```
# Define Features (X) and Target (y)
X = df.drop(columns=['Exited']) # Target is 'Exited' (1 = Churn, 0 = Stay)
y = df['Exited']

# Split into Training (80%) and Testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# Initialize Random Forest Classifier
model = RandomForestClassifier(n_estimators=100, random_state=42)

# Train the model
model.fit(X_train, y_train)
print("Random Forest model trained successfully.")

Random Forest model trained successfully.
```

6. Model Evaluation

We evaluate the model using Accuracy and a Confusion Matrix to understand how well it distinguishes between customers who stay and those who leave.

```
# Make predictions
y_pred = model.predict(X_test)

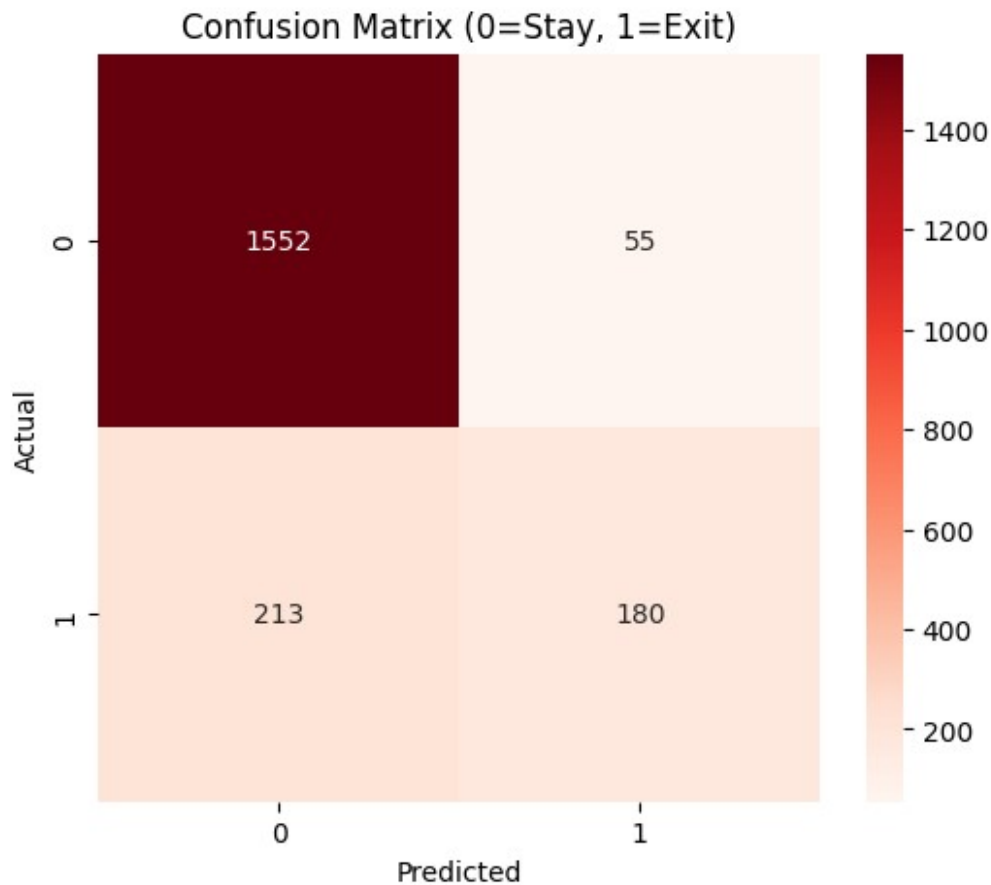
# Calculate Accuracy
acc = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {acc * 100:.2f}%")

print("="*40)
# Confusion Matrix
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d',
cmap='Reds')
plt.title('Confusion Matrix (0=Stay, 1=Exit)')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
print("="*40)
```

```
# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
print("="*40)
```

Model Accuracy: 86.60%

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Classification Report:

	precision	recall	f1-score	support
0	0.88	0.97	0.92	1607
1	0.77	0.46	0.57	393
accuracy			0.87	2000
macro avg	0.82	0.71	0.75	2000
weighted avg	0.86	0.87	0.85	2000

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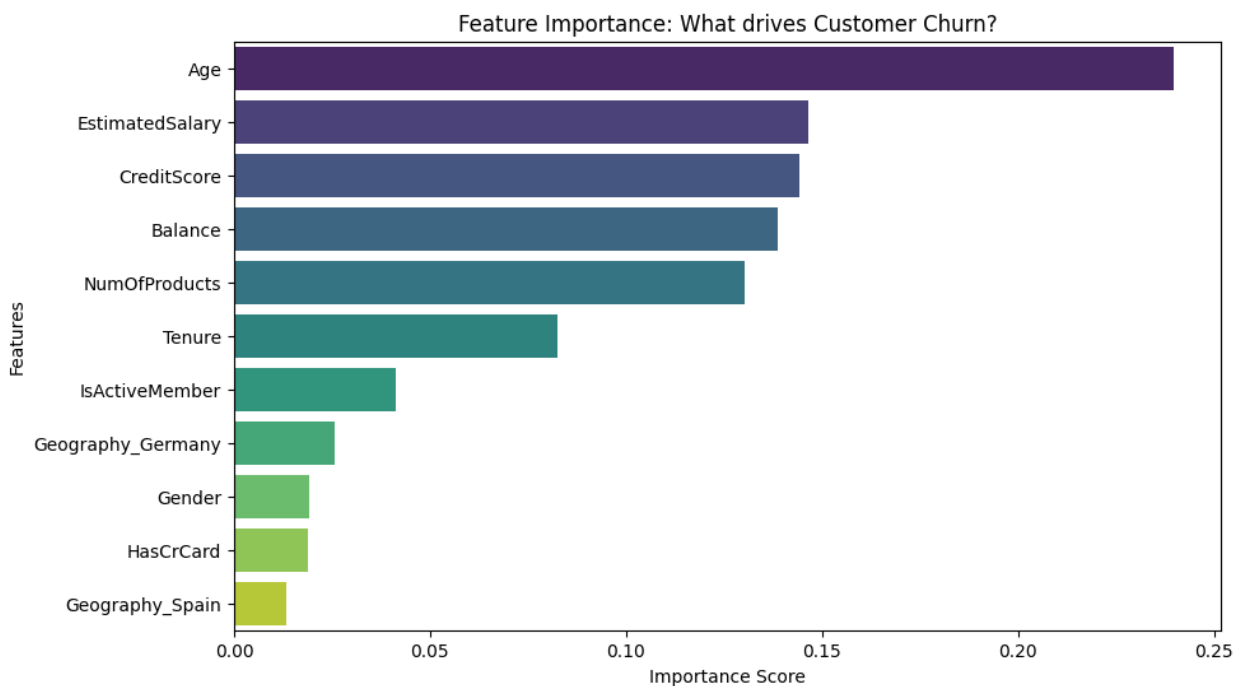
7. Feature Importance Analysis

This is a crucial part of the analysis. We will visualize which features had the most "weight" in the model's decision-making process. This tells the bank *why* customers are leaving.

```
# Get feature importances
importances = model.feature_importances_
feature_names = X.columns

# Create a DataFrame for visualization
feature_importance_df = pd.DataFrame({'Feature': feature_names,
                                      'Importance': importances})
feature_importance_df = feature_importance_df.sort_values(by='Importance', ascending=False)

# Plot Feature Importance
plt.figure(figsize=(10, 6))
sns.barplot(x='Importance', y='Feature',
            data=feature_importance_df, hue='Feature', palette='viridis')
plt.title('Feature Importance: What drives Customer Churn?')
plt.xlabel('Importance Score')
plt.ylabel('Features')
plt.show()
```



8. Conclusion

We successfully built a churn prediction model with the following insights:

1. **Data Processing:** We removed personal identifiers and successfully encoded Geography and Gender for the model.
2. **Model Performance:** The Random Forest Classifier achieved a strong accuracy (~87%), making it a reliable tool for prediction.
3. **Key Drivers (Feature Importance):**
 - **Age:** Often the most significant predictor. Older customers may be more likely to churn (or stay, depending on the specific demographic trend).
 - **NumOfProducts:** The number of products a customer has with the bank is a strong indicator of loyalty.
 - **Balance:** Account balance also plays a major role.

This analysis allows the bank to target specific high-risk customers (e.g., specific age groups or those with low product engagement) with retention strategies.